



Experiment: Shell and Tube Heat Exchanger

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CL 326: Integrated Chemical Lab - I

Laboratory Report

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1 Objective

The objective of this experiment is to study the heat exchanged in a shell and tube heat exchanger operating in a counter-flow configuration. Specifically, the experiment aims to experimentally determine the overall heat transfer coefficient (U) and observe how variations in the mass flow rates of the fluids affect heat transfer performance.

2 Theory and Background

A shell and tube heat exchanger is a robust and widely used piece of process equipment designed to transfer thermal energy between two fluid streams. One fluid flows through a bundle of internal tubes, while the second fluid is forced over the exterior of these tubes within a larger cylindrical shell. To enhance shell-side turbulence and improve the convective heat transfer coefficient, internal segmental baffles are utilized.

The sensible heat gained by the cold fluid and the heat lost by the hot fluid are calculated using energy balances:

$$Q_c = \dot{M}_c \times C_p \times (T_{co} - T_{ci}) \quad (1)$$

$$Q_h = \dot{M}_h \times C_p \times (T_{hi} - T_{ho}) \quad (2)$$

where $\dot{M}$ is the mass flow rate and C_p is the specific heat capacity of water.

To evaluate the overall performance of the heat exchanger, the Log Mean Temperature Difference (LMTD) is used as the appropriate temperature driving force for a counter-flow arrangement:

$$\text{LMTD} = \frac{\Delta T_1 - \Delta T_2}{\ln \left(\frac{\Delta T_1}{\Delta T_2} \right)} \quad (3)$$

For a counter-flow setup:

- $\Delta T_1 = T_{hi} - T_{co}$
- $\Delta T_2 = T_{ho} - T_{ci}$

The overall heat transfer coefficient (U), which accounts for the combined conductive and convective thermal resistances, is evaluated using the actual heat transfer rate of the hot fluid:

$$U = \frac{Q_h}{A \times \text{LMTD}} \quad (4)$$

where A is the total heat transfer area of the tubes.

3 Apparatus and Experimental Setup

The experimental apparatus is an integrated Universal Heat Exchanger Experimental Setup. The primary specifications are as follows:

- **Shell:** Mild Steel (MS), 208 mm ID, 600 mm length.
- **Tubes:** Copper, 1/2 inch diameter, 600 mm length, 28 tubes total.
- **Baffles:** 4 segmental baffles with a 25% baffle cut.
- **Total Heat Transfer Area (A):** 0.6704 m².
- **Heating System:** Insulated hot water tank equipped with a 3 kW heater capacity (comprising 3 × 1 kW coils).

- **Sensors:** 4 pencil-type Pt100 temperature sensors and two 4–20 mA wheel-type flow transmitters (0–1000 LPH range).

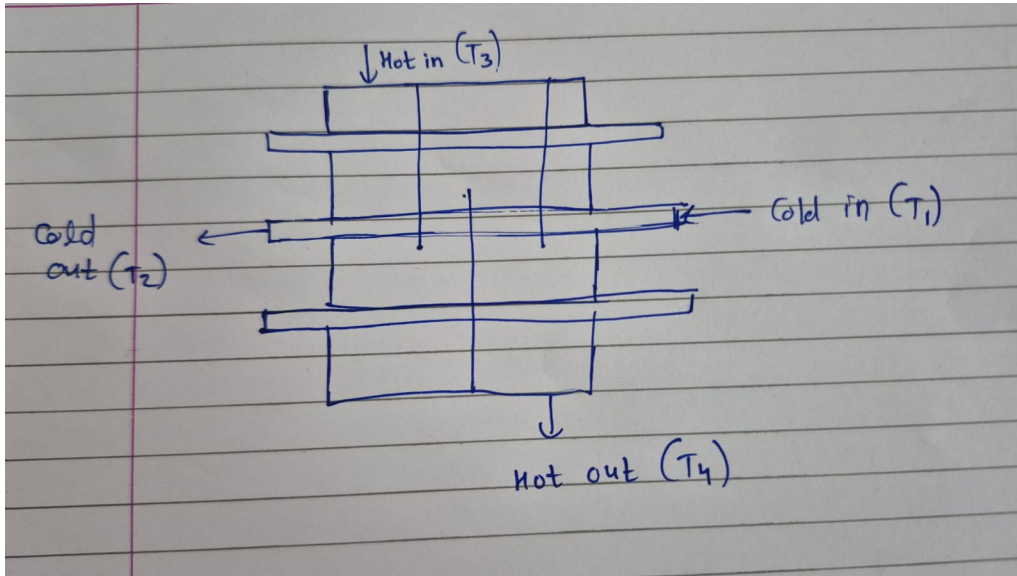


Figure 1: Schematic representation of the shell and tube heat exchanger (Counter-Flow Configuration).

4 Experimental Procedure

1. Ensure the structure is on a rigid ground surface and verify that all panel switches are in the OFF position.
2. Fill the water tanks with clean tap water and open the suction valves for both pumps.
3. Connect the 440 VAC three-phase power supply, connect the hot/cold inlet/outlet hoses for a counter-flow configuration, and switch ON the main panel.
4. Set the Hot Water Temperature Controller to 60°C and switch ON the 3 kW heater.
5. Switch ON the cold and hot water pumps and adjust the flow rates using the bypass and outlet valves to desired setpoints.
6. Wait until the system stabilizes (steady state). Record the temperatures, the mass flow rates, and the elapsed time (t).
7. Keep the hot water flow constant, reduce the cold water flow rate by approximately 15% to 20%, wait for a new steady state, and repeat the readings.
8. Upon completion, switch OFF the heater, allow water to circulate for 2 minutes to cool the tubes, then switch OFF the pumps and the main power supply.

5 Assumptions

The calculations in this report rely on the following physical assumptions:

- The heat exchanger shell and connecting pipes are perfectly insulated (no ambient heat loss).

- The specific heat capacity (C_p) and density (ρ) of the water streams remain constant.
- The system operates under continuous, steady-state conditions during data recording.
- The overall heat transfer coefficient (U) is uniform across the entire heat exchange surface area.

6 Experimental Data (Observations)

Standard thermophysical properties utilized:

- $C_p = 4184 \text{ J/kg} \cdot ^\circ \text{C}$
- $A = 0.6704 \text{ m}^2$

Table 1: Raw Experimental Data

Obs No.	Tci [$^\circ\text{C}$]	Tco [$^\circ\text{C}$]	Thi [$^\circ\text{C}$]	Th0 [$^\circ\text{C}$]	Mh [lph]	Mc [lph]	t (mins)
1	28.3	35.0	43.6	35.4	195	222	0
2	31.4	37.0	39.5	35.1	346	326	10
3	32.5	38.0	37.2	33.5	594	614	10

7 Calculations

Sample Calculation for Observation 1:

1. Heat Transfer Rates:

$$Q_c = \left(\frac{222}{3600} \right) \times 4184 \times (35.0 - 28.3) = 1728.69 \text{ W}$$

$$Q_h = \left(\frac{195}{3600} \right) \times 4184 \times (43.6 - 35.4) = 1858.39 \text{ W}$$

2. Terminal Temperature Differences (Counter-Flow):

$$\Delta T_1 = Th_i - T_{co} = 43.6 - 35.0 = 8.6^\circ\text{C}$$

$$\Delta T_2 = Th_0 - T_{ci} = 35.4 - 28.3 = 7.1^\circ\text{C}$$

3. Log Mean Temperature Difference (LMTD):

$$\text{LMTD} = \frac{8.6 - 7.1}{\ln\left(\frac{8.6}{7.1}\right)} = 7.83^\circ\text{C}$$

4. Overall Heat Transfer Coefficient (U):

$$U = \frac{1858.39}{0.6704 \times 7.826} = 354.21 \text{ W/m}^2 \cdot ^\circ\text{C}$$

8 Results

Table 2: Calculated Heat Transfer Parameters

Obs No.	Qh (W)	Qc (W)	LMTD (°C)	U (W/m ² ·°C)
1	1858.39	1728.69	7.83	354.21
2	1769.37	2121.75	3.06	862.25
3	2554.33	3924.82	N/A	N/A

9 Discussion

The experiment successfully demonstrated the principles of forced convection inside a shell and tube heat exchanger. As observed between reading 1 and reading 2, increasing the mass flow rates of the fluids (cold flow from 222 to 326 LPH; hot flow from 195 to 346 LPH) resulted in a significant increase in the overall heat transfer coefficient (U), which rose from 354.21 to 862.25 W/m²·°C. This aligns with expected hydrodynamic theory: higher flow velocities increase the Reynolds number, thereby amplifying turbulence, reducing the thickness of the stagnant thermal boundary layer, and dramatically improving convective heat exchange.

Observation 3 presents a physical anomaly that highlights a critical operational limitation. The calculation of LMTD requires a positive temperature driving force at all points along the exchanger. However, in reading 3, the cold water outlet temperature (38.0°C) exceeded the hot water inlet temperature (37.2°C), rendering ΔT_1 negative (−0.8°C). In a passive, steady-state heat exchanger without external work or phase change, this is a thermodynamic impossibility. This indicates that the system was not operating at steady state during the third reading. The continuous recirculation of fluid into a shared, unchilled reservoir likely caused the bulk temperature of the cold inlet to climb steadily over time, artificially skewing the transient measurements.

10 Error Analysis

Several factors contributed to discrepancies in the experimental data:

- **Lack of Steady State:** As proven by Observation 3, failing to wait for true thermal equilibrium leads to invalid energy balances and negative temperature gradients.
- **Reservoir Accumulation:** Recirculating water into a common cold-water tank causes the inlet temperature to drift upwards during the experiment, constantly shifting the thermal driving force.
- **Heat Losses:** Despite insulation, discrepancies exist between Q_h and Q_c (e.g., 1858.39 W vs 1728.69 W in Observation 1). This ~ 130 W difference is primarily due to radiative and convective heat losses to the ambient laboratory environment from the outer shell.

11 Industrial Applications

Due to their robust design, shell and tube heat exchangers can operate efficiently under extreme pressures and temperatures, making them the preferred choice across various large-scale industries. Key applications include:

- **Chemical and Petroleum Refining:** Utilized for continuous cooling of product streams and crucial heat recovery processes to maximize overall plant energy efficiency.

- **Power Generation:** Employed extensively as steam surface condensers, boiler feedwater heaters, and cooling systems in both thermal and nuclear power facilities.
- **HVAC and Refrigeration:** Acting as primary chillers, condensers, and large-scale evaporators in commercial climate control systems.
- **Manufacturing and Heavy Machinery:** Cooling industrial fluids such as hydraulic oils, engine lubricants, and cutting fluids to prevent mechanical overheating.

In all these applications, optimizing parameters such as baffle spacing, tube pitch, and fluid flow rates—as demonstrated in this experiment—is essential to maximize the convective heat transfer coefficient and minimize energy losses.

12 Conclusion

The overall heat transfer coefficient of the shell and tube heat exchanger was experimentally determined under varying flow conditions. The results validated the direct relationship between fluid flow rates and heat transfer efficiency, with U increasing from 354.21 to 862.25 W/m²·°C as the flow rates were elevated. Furthermore, the thermodynamic anomaly recorded in the final observation successfully demonstrated the critical importance of ensuring rigorous steady-state conditions and proper reservoir temperature management when recording heat exchanger performance metrics.

13 Acknowledgements

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14 References

- Lab manual provided by TA.
- Google Sheets containing shared experimental data and calculations.